Exercise 3F

Understanding

1 Simplify using index form.

a $3 \times 3 \times 3 \times 3$ b $7 \times 7 \times 7 \times 7 \times 7 \times 7$
c $8 \times 8 \times 8$ d $2 \times x \times x \times x \times 3 \times x$
e $3 \times y \times y \times 5 \times y \times y$ f $2 \times b \times a \times 4 \times b \times a \times a$

2 Copy and complete this table.

x	4	3	2	1	0
2^x		$2^3 = 8$			

3 Copy and complete.

a $2^2 \times 2^3 = 2 \times 2 \times$ ______
$= 2^{-}$

b $\dfrac{x^5}{x^3} = \dfrac{x \times x \times x \times x \times \_\_\_\_}{\_\_\_\_\_\_}$
$= x^{-}$

c $(a^2)^3 = a \times a \times$ ______ $\times$ ______
$= a^{-}$

d $(2x)^0 \times 2x^0 =$ ___ $\times$ ___
$= 2$

Fluency

Example 13 4 Simplify using the first index law.

a $a^5 \times a^4$ b $x^3 \times x^2$ c $b \times b^5$
d $7m^2 \times 2m^3$ e $2s^4 \times 3s^3$ f $t^8 \times 2t^8$
g $\frac{1}{5}p^2 \times p$ h $\frac{1}{4}c^4 \times \frac{2}{3}c^3$ i $\frac{3}{5}s \times \frac{3s}{5}$
j $2x^2y \times 3xy^2$ k $3a^2b \times 5ab^5$ l $3v^7w \times 6v^2w$
m $3x^4 \times 5xy^2 \times 10y^4$ n $2rs^3 \times 3r^4s \times 2r^2s^2$ o $4m^6n^7 \times mn \times 5mn^2$

Example 14 5 Simplify using the second index law.

a $x^5 \div x^2$ b $a^7 \div a^6$ c $q^9 \div q^6$
d $b^5 \div b$ e $\dfrac{y^8}{y^3}$ f $\dfrac{d^8}{d^3}$
g $\dfrac{j^7}{j^6}$ h $\dfrac{m^{15}}{m^9}$ i $2x^2y^3 \div x$
j $3r^5s^2 \div (r^3s)$ k $6p^4q^2 \div (3q^2p^2)$ l $16m^7x^5 \div (8m^3x^4)$
m $\dfrac{5a^2b^4}{a^2b}$ n $\dfrac{8st^4}{2t^3}$ o $\dfrac{2v^5}{8v^3}$
p $\dfrac{7a^2b}{14ab}$ q $\dfrac{-3x^4y}{9x^3y}$ r $\dfrac{-8x^2y^3}{16x^2y}$

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Example 15 6 Simplify using the third, fourth and fifth index laws.

a $(x^5)^2$ b $(t^3)^2$ c $4(a^2)^3$ d $5(y^5)^3$
e $(4t^2)^3$ f $(2u^2)^2$ g $(3r^3)^3$ h $(3p^4)^4$
i $\left(\dfrac{a^2}{b^3}\right)^2$ j $\left(\dfrac{x^3}{y^4}\right)^3$ k $\left(\dfrac{x^2y^3}{z^4}\right)^2$ l $\left(\dfrac{u^4w^2}{v^2}\right)^4$
m $\left(\dfrac{3f^2}{5g}\right)^3$ n $\left(\dfrac{3a^2b}{2pq^3}\right)^2$ o $\left(\dfrac{at^3}{3g^4}\right)^3$ p $\left(\dfrac{4p^2q^3}{3r}\right)^4$

Example 16 7 Evaluate the following using the zero power.

a $8x^0$ b $3t^0$ c $(5z)^0$ d $(10ab^2)^0$
e $5(g^3h^3)^0$ f $8x^0 - 5$ g $4b^0 - 9$ h $7x^0 - 4(2y)^0$

Problem-solving

8 Use appropriate index laws to simplify the following.

a $x^6 \times x^5 \div x^3$ b $x^2y \div (xy) \times xy^2$
c $x^4n^7 \times x^3n^2 \div (xn)$ d $\dfrac{x^2y^3 \times x^2y^4}{x^3y^5}$
e $\dfrac{m^2w \times m^3w^2}{m^4w^3}$ f $\dfrac{r^4s^7 \times r^4s^7}{r^4s^7}$
g $\dfrac{9x^2y^3 \times 6x^7y^5}{12xy^6}$ h $\dfrac{4x^2y^3 \times 12x^2y^2}{24x^4y}$
i $\dfrac{16a^8b \times 4ab^7}{32a^7b^6}$ j $(3m^2n^4)^3 \times mn^2$
k $-5(a^2b)^3 \times (3ab)^2$ l $(4f^2g)^2 \times f^2g^4 \div (3(fg^2)^3)$
m $\dfrac{4m^2n \times 3(m^2n)^3}{6m^2n}$ n $\dfrac{(7y^2z)^2 \times 3yz^2}{7(yz)^2}$
o $\dfrac{2(ab)^2 \times (2a^2b)^3}{4ab^2 \times 4a^7b^3}$ p $\dfrac{(2m^3)^2}{3(mn^4)^0} \times \dfrac{(6n^5)^2}{(-2n)^3m^4}$

9 Simplify:

a $(-3)^3$ b $-(3)^3$ c $(-3)^4$ d -3^4

10 Simplify:

a $\left((x^2)^3\right)^2$ b $\left((a^5)^3\right)^7$ c $\left(\left(\dfrac{a^2}{b}\right)^3\right)^5$

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Reasoning

11 Evaluate without the use of a calculator.

a $\frac{13^3}{13^2}$ b $\frac{18^7}{18^6}$ c $\frac{9^8}{9^6}$ d $\frac{4^{10}}{4^7}$

e $\frac{25^2}{5^4}$ f $\frac{36^2}{6^4}$ g $\frac{27^2}{3^4}$ h $\frac{32^2}{2^7}$

12 When Billy uses a calculator to raise -2 to the power 4 he gets -16 when the answer is actually 16. What has he done wrong?

13 Find the value of a in these equations in which the index is unknown.

a $2^a = 8$ b $3^a = 81$ c $2^{a+1} = 4$

d $(-3)^a = -27$ e $(-5)^a = 625$ f $4^a = \frac{1}{4}$

Enrichment: Indices in equations

14 If $x^4 = 3$ find the value of:

a x^8 b $x^4 - 1$ c $2x^{16}$ d $3x^4 - 3x^8$

15 Find the value(s) of x.

a $x^4 = 16$ b $2^{x-1} = 16$ c $2^{2x} = 16$ d $2^{2x-3} = 16$

16 Find the possible pairs of positive integers for x and y when:

a $x^y = 16$ b $x^y = 64$ c $x^y = 81$ d $x^y = 1$

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14 a $\frac{5\sqrt{3}-5}{2}$ b $2\sqrt{3}+2$ c $3\sqrt{5}+6$

d $-4-4\sqrt{2}$ e $\frac{-3-3\sqrt{3}}{2}$ f $\frac{42+7\sqrt{7}}{29}$

g $-12-4\sqrt{10}$ h $-14-7\sqrt{5}$ i $\frac{2\sqrt{11}+2\sqrt{2}}{9}$

j $2\sqrt{5}-2\sqrt{2}$ k $\sqrt{7}-\sqrt{3}$ l $\frac{\sqrt{14}-\sqrt{2}}{6}$

m $\frac{6+\sqrt{6}}{5}$ n $\sqrt{14}+2\sqrt{2}$ o $10-4\sqrt{5}$

p $\frac{b\sqrt{a}-b\sqrt{b}}{a-b}$ q $\frac{a\sqrt{a}+a\sqrt{b}}{a-b}$ r $\frac{a+b-2\sqrt{ab}}{a-b}$

s $\frac{a-\sqrt{ab}}{a-b}$ t $\frac{a\sqrt{b}+b\sqrt{a}}{a-b}$

Exercise 3F

1 a 3^4 b 7^6 c 8^3

d $6x^3$ e $15y^4$ f $8a^3b^2$

2

x	4	3	2	1	0
$2x$	$2^4=16$	$2^3=8$	$2^2=4$	$2^1=2$	$2^0=1$

3 a $2^2\times2^3=2\times2\times2\times2\times2$
$=2^5$

b $\frac{x^5}{x^3}=\frac{x\times x\times x\times x\times x}{x\times x\times x}$
$=x^2$

c $(a^2)^3=a\times a\times a\times a\times a\times a$
$=a^6$

d $(2x)^0\times2x^0=1\times2$
$=2$

4 a a^9 b x^5 c b^6

d $14m^5$ e $6s^7$ f $2t^{16}$

g $\frac{p^3}{5}$ h $\frac{c^7}{6}$ i $\frac{9}{25}s^2$

j $6x^3y^3$ k $15a^3b^6$ l $18v^9w^2$

m $150x^5y^6$ n $12r^7s^5$ o $20m^8n^{10}$

5 a x^3 b a c q^3 d b^4

e y^5 f d^5 g j h m^6

i $2xy^3$ j $3r^2s$ k $2p^2$ l $2m^4x$

m $5b^3$ n $4st$ o $\frac{1}{4}v^2$ p $\frac{1}{2}a$

q $-\frac{x}{3}$ r $-\frac{y}{2}$

6 a x^{10} b t^6 c $4a^6$ d $5y^{15}$

e $64t^6$ f $4u^4$ g $27r^9$ h $81p^{16}$

i $\frac{a^4}{b^6}$ j $\frac{x^9}{y^{12}}$ k $\frac{x^4y^6}{z^8}$ l $\frac{u^{16}w^8}{v^8}$

m $\frac{27f^6}{125g^3}$ n $\frac{9a^4b^2}{4p^2q^8}$ o $\frac{a^3t^9}{27g^{12}}$ p $\frac{256p^8q^{12}}{81r^4}$

7 a 8 b 3 c 1 d 1

e 5 f 3 g -5 h 3

8 a x^8 b x^2y^2 c x^6n^8

d xy^2 e m f r^4s^7

g $\frac{9x^8y^2}{2}$ h $2y^4$ i $2a^2b^2$

j $27m^7n^{14}$ k $-45a^8b^5$ l $\frac{16}{3}f^3$

m $2m^6n^3$ n $21y^3z^2$ o 1 p $-6m^2n^7$

9 a -27 b -27 c 81 d -81

10 a x^{12} b a^{105} c $\frac{a^{30}}{b^{15}}$

11 a 13 b 18 c 81 d 64

e 1 f 1 g 9 h 8

12 He has not included the minus sign inside the brackets, i.e. has only applied it afterwards. Need $(-2)^4$ not -2^4.

13 a 3 b 4 c 1

d 3 e 4 f -1

14 a 9 b 2 c 162 d -18

15 a ±2 b 5 c 2 d $\frac{7}{2}$

16 a $x=2, y=4$ or $x=4, y=2$ or $x=16, y=1$

b $x=8, y=2$ or $x=4, y=3$ or $x=64, y=1$, or $x=2, y=6$

c $x=9, y=2$, or $x=3, y=4$ or $x=81, y=1$

d $x=1, y=$ any integer

Exercise 3G

1 a $2^{-2}, 2^{-3}, 2^{-4}$ b x^{-1}, x^{-2}, x^{-3} c $3x^{-1}, 3x^{-2}, 3x^{-3}$

2 a $\frac{1}{3^2}$ b $\frac{1}{5^2}$ c $\frac{5}{4^2}$ d $\frac{-2}{3^3}$

3 a $\frac{1}{a^b}$ b a^b

4 a $\frac{1}{x^5}$ b $\frac{1}{a^4}$ c $\frac{2}{m^4}$ d $\frac{3}{y^7}$

e $\frac{3a^2}{b^3}$ f $\frac{4m^3}{n^3}$ g $\frac{10y^5z}{x^2}$ h $\frac{3z^3}{x^4y^2}$

i $\frac{q^3r}{3p^2}$ j $\frac{d^2f^5}{5e^4}$ k $\frac{3u^2w^7}{8v^6}$ l $\frac{2b^3}{5c^5d^2}$

5 a x^2 b $2y^3$ c $4m^7$ d $3b^5$

e $2b^4d^6$ f $3m^2n^4$ g $\frac{4b^4a^3}{3}$ h $\frac{5h^3g^3}{2}$

6 a x b a^3 c $\frac{2}{b^4}$ d $\frac{3}{y^3}$

e $\frac{6}{a^2}$ f $\frac{12}{x}$ g $\frac{-10}{m^6}$ h $\frac{-18}{a^{10}}$

i $\frac{2x}{3}$ j $\frac{7d^2}{10}$ k $\frac{1}{2}$ l $\frac{3b^2}{4}$

m $\frac{5}{3s^3}$ n $\frac{4}{3f^2}$ o $\frac{1}{2d^2}$ p $\frac{5}{6t^2}$

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